Proposing a Classification Scheme for AI Usage in Scientific Articles

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Abstract

Journals and preprint servers increasingly ask authors to disclose their use of artificial intelligence (AI), but such disclosures are almost always free text that varies from journal to journal and cannot be searched or compared. We propose AIL2026 (AI Labels), a compact, machine-readable classification of AI usage. Like the CRediT taxonomy for human contributors, it records who contributed what; like MSC2020, it uses short standardized codes, but with plain-word labels that humans can read at a glance. Each paper carries a tag: a headline level from `AI-0` to `AI-4` summarizing AI's role in the research itself, followed by labels stating where AI contributed and how much, each with a depth from 0 to 4. Examples are `AI-0: Text1 Check2` and `AI-3: Proof3 Text1`. The same labels apply to every arXiv category, with field-specific interpretations. AI can contribute to research in very different ways, from light uses such as polishing text or double-checking signs and prefactors to substantial contributions such as suggesting research directions or proving theorems. We argue that light use and AI verification of any depth benefit everyone and should be encouraged: polishing makes papers easier to read, and checks with Python or Lean add a further layer of verification. Normalizing light use, which is already widespread but often undisclosed, also lowers the barrier to disclosing heavier use, since transparency, not the level of use, is what signals scientific integrity. Where AI contributes essential parts of the research, disclosure is required by good scientific practice: undisclosed use at that level confers an unfair advantage. We discuss design principles, edge cases, incentives, and a route to adoption beginning with a pilot in mathematics.

AI usage classification (AIL2026). `AI-3: Idea3 Result3 Proof3 Text3 Lit2 Figs3 Check2 [Claude Opus 5.5, 2026-10]` (see the final section).

Mathematics Subject Classification (MSC2020). Primary 00A35; Secondary 68T01.

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1 Introduction

AI systems are now used at every stage of research: polishing prose, searching the literature, writing code, analyzing data, suggesting conjectures, and, increasingly, producing proofs. Their impact has been recognized at the highest level: the 2024 Nobel Prize in Chemistry was awarded in part for AI-based protein structure prediction [1]. In 2026 the pace accelerated further. An Anthropic model computed a nine-loop scattering amplitude in planar $\mathcal{N} = 4$ super-Yang–Mills theory with little human guidance beyond the instruction to keep going, a result independently validated by a leading expert [2]. OpenAI announced a resolution of the Navier–Stokes Millennium Prize Problem [3]: a proof of finite-time blow-up, produced by an internal multi-agent system steered by its researchers and accompanied by a Lean formalization [4]. Concurrently, Alpöge and Buckmaster obtained finite-time blow-up for the Euler equations with smooth forcing, with proofs generated by several AI models under their direction [5]. In October, OpenAI released more than 700 mathematical manuscripts produced by an internal model, with Lean formalizations for part of them [6]. Even for these results, the extent of human guidance and the possible influence of earlier inputs were questioned after the fact [4, 5], which shows how hard it is to reconstruct who contributed what once a result is public.

These developments put pressure on the publication system. Submissions to arXiv have doubled within two years, to over 40,000 in September 2026, and arXiv now limits each submitter to two submissions per month, naming AI tools as one driver of this growth [7]. Publishers and preprint servers have responded by asking authors to disclose AI use, and new venues such as the Hexagon repository [8] attach labels of their own. In practice, most disclosures are paragraphs of free text placed in acknowledgments or dedicated statements. They differ in granularity, they cannot be filtered or aggregated, and a reader cannot tell at a glance whether AI fixed the grammar or proved the main theorem.

Fields have long solved an analogous problem for *subject matter*. The Mathematics Subject Classification (MSC2020) [9] and the ACM Computing Classification System [10] give every paper short, standardized codes that are displayed in metadata and indexed by search tools. For contributions rather than subjects, the CRediT taxonomy [11] plays a similar role for human contributors, but it does not cover AI. We argue that AI usage deserves the same treatment, with one important difference: subject codes such as `00A35` and `68T01`, the classes of this paper, are meaningful only to those who have looked them up (here, “methodology of mathematics”

and "general topics in artificial intelligence"), whereas a statement about AI use should be understood immediately by any reader, referee or editor. We therefore propose AIL2026 (*AI Labels*): a short tag built from plain words and small numbers, attached to the paper's metadata, that says what AI contributed and where. As with MSC2020, the year identifies the version of the definitions, which will need revision as AI capabilities change.

This paper makes four contributions:

1. an argument that light AI use benefits the scientific record and should be encouraged, together with a principled line beyond which disclosure becomes essential (Section 3);
2. a set of design principles for such a classification (Section 4);
3. a concrete scheme of universal, self-explanatory labels with a depth scale and a headline content level, together with brief glosses for individual fields (Sections 5–6);
4. a discussion of limitations (Section 7), a comparison with existing proposals (Section 8), and a path to adoption (Section 9).

The proposal has a fixed core and adjustable details. The core consists of four elements: a short tag of plain-word labels, each with a depth from 0 to 4; a headline content level summarizing AI's role in the research itself; the encouragement of light, beneficial AI use; and a line, drawn on grounds of fairness, beyond which AI contributions must be disclosed. The details, such as the exact set of labels, the wording of the depth tests, and the field glosses, are open to refinement in discussion with the communities that would use the scheme.

2 Motivation

The scheme serves several groups, but one purpose underlies them all: the credibility of science. Readers can trust published work only if they know how it was produced.

Readers and referees. The appropriate level of scrutiny depends on what AI did. A paper whose prose was polished by AI needs no special attention; a paper whose central proof was generated by AI may warrant a closer check of every step, or evidence of formal verification.

Editors and policy. Journals and preprint servers are writing policies on AI use. A shared vocabulary makes such policies precise ("submissions tagged `Proof4` must include a formal certificate") and allows the effect of a policy to be measured. As submission volumes grow [7], a tag that can be read at a glance may also help moderators and editors handle them consistently.

Meta-science. Standardized tags let researchers study how AI use is changing across fields and over time, how it correlates with retractions or corrections, and where it is having the greatest effect on discovery.

Credit. Making AI contributions explicit also makes *human* contributions explicit. A tag stating that the ideas and proofs are human, with AI used only for language, is a positive signal, not merely the absence of one. For authors who use no AI at all, the bare tag `AI-0` states this at a glance.

3 Encouraged use and the line that matters

Discussion of AI in research often treats any use as suspect, and assumes that AI assistance saves authors work. Often it does, but not always: an author who has AI check a finished draft and then works through the resulting corrections may spend more time on the paper, not less. We take a different view of AI use: some uses make papers better for everyone who reads them and therefore should be actively encouraged, while others change who did the research and must be disclosed. A classification scheme is useful only if it distinguishes the two.

3.1 Light use improves the scientific record

Many errors that survive into published papers are not errors of reasoning but of presentation and bookkeeping. They cost readers time, can mislead those who reuse results, and are exactly the kind of error that referees, reading for substance, are least likely to catch. Current AI tools are well suited to finding them. Concrete examples include:

- **Language.** Typos, spelling and grammar mistakes, ambiguous sentences, and inconsistent terminology, which particularly burden authors writing in a second language and readers doing the same.
- **LaTeX cross-references.** A reference to "equation (3.7)" that should point to (3.8) after a revision, an `\eqref` to the wrong label, or a broken `\ref` that appears as "(??)" in the compiled PDF.
- **Citations.** A standard result used without attribution, a citation to the wrong theorem number in the cited work, a bibliography entry with an incorrect year or journal, or a missing reference to closely related prior work.
- **Notation.** A symbol defined one way in Section 2 and used differently in Section 5, or a hypothesis invoked in a proof but absent from the theorem statement.
- **Copy-and-paste errors.** A paragraph duplicated during editing, a formula carried over from another section with outdated variable names, a duplicated LaTeX label that makes references point to the wrong equation, or a lemma copied from an earlier paper together with hypotheses that no longer apply.
- **Computational slips.** A sign error in an intermediate identity, a dropped factor of $\frac{1}{2}$ or 2π, an index shifted by one, or inconsistent units, each of which may propagate into the work of anyone who builds on the result.

None of these changes the scientific contribution of the paper, and every correction is checked and accepted by the authors. Yet each one makes the paper easier to read, quicker to referee, and safer to build on. In the scheme of Section 5 they are recorded under the labels `Text`, `Lit` and `Check` at low depth, for example `AI-0: Text1 Check2`. The headline `AI-0` tells the reader at once that the intellectual content of the paper remains entirely human.

3.2 An additional layer of verification

Peer review is the community's main safeguard, but it is under strain. Submission numbers are rising sharply: arXiv's doubled within two years, straining even its moderation, which is far lighter than peer review [7]. Referees are unpaid, time-limited, and rightly concentrate on the correctness and significance of the main results. Detailed checking of every computation, cross-reference and citation is rarely possible. Referees are often asked not to upload manuscripts to AI tools, for confidentiality and to keep unpublished results out of training data, so AI checking has to come from the authors. An author who asks an AI tool to re-derive key identities, check each equation reference, and compare the bibliography against the text before submission adds a layer of verification even before peer review begins. This complements the referee rather than replacing them: the referee can spend their limited time on the mathematics, not on typographical bookkeeping. Readers of the preprint, who often rely on its results long before the final journal version appears, benefit as well.

AI checking is not infallible. It may miss errors or report ones that do not exist, so the authors must confirm every proposed correction themselves. Provided its suggestions are reviewed in this way, an additional independent check, even an imperfect one, catches errors that would otherwise be missed at little risk of introducing new ones. For this reason we regard `Check` as a positive attribute at every depth: a paper reporting AI verification of its computations tells readers that an extra check was performed. Even `Check4`, for instance a complete formal verification produced largely by AI, is an addendum that strengthens the paper, and authors should publish it as supplementary material, so that readers can check it.

3.3 Disclosure without stigma

If a classification is perceived as a mark against a paper, authors will under-report, as people tend to conceal an attribute that discredits them rather than disclose it [12], and the scheme will fail at its basic task. Light, beneficial use must therefore be visibly treated as good practice. Reporting `Text1` or `Check2` should carry no more stigma than thanking a colleague for proofreading, and editors and referees should be explicitly asked not to penalize it. As noted above, light use is not the same as outsourcing work: a tag such as `Text1` or `Check2` records that a tool was involved, not that the author did less of the work. A scheme that rewards honesty about light use also makes honesty about heavier use more likely: once reporting light use is routine, reporting `Proof3` is just one more step on the same scale rather than a confession. Public discussion of AI use is currently highly moralized, and its judgment is changing quickly; a scheme that records provenance rather than passing judgment remains useful as these norms shift.

3.4 Where the line lies

The original motivation for this proposal is fairness and transparency. When AI does essential parts of the research, such as formulating the main result, supplying the key idea of a proof, designing the experiment, or choosing the analysis, a scientist who uses it without saying so gains an advantage over colleagues who do the same work themselves. Their output appears to reflect their own ability, and comparisons for hiring, funding, and promotion become distorted. Readers also lose information they need to judge how the result was obtained and how carefully it should be checked. Heavy AI use in supporting tasks, such as writing up or coding, gives a smaller advantage of the same kind, in time saved; the scheme makes it visible too, at lower levels (Section 5).

We therefore draw the line at content level `AI-3`: depth 3 or more in any of the labels that carry the intellectual content of a paper, namely `Idea`, `Method`, `Result` and `Proof` (Table 1). Below that line, AI use is assistance and is encouraged. At or above it, AI is contributing essential parts of the work, and disclosure is not a courtesy but a requirement of scientific honesty. Such use is not wrong in itself: the results may be correct, valuable, and well verified. What is wrong is presenting them as unassisted.

Table 1: How different uses of AI should be viewed. Labels and depths are defined in Tables 2 and 3.

Zone	Typical tags	Examples
Encouraged	`AI-0` with `Text1`, `Lit1`–`Lit2`, `Figs1`, `Check1`–`Check4`	Proofreading; fixing broken references and citations; double-checking computations; literature search; tidying figures.
Neutral, report	`AI-0` to `AI-2`, e.g. `Proof1`–`Proof2`, `Text2`–`Text3`, `Code2`, `Figs2`	Drafting background sections or routine code; streamlining a human proof without changing its core argument.
Essential, must disclose	`AI-3` or `AI-4`: `Idea`, `Method`, `Result` or `Proof` at depth 3–4	AI-formulated conjectures; AI-strengthened theorems; AI-derived proofs or analyses; AI-designed experiments.

4 Design principles

Readable at a glance.
A tag should be understood without a lookup table. Labels are short English words, not letters or numbers, and every tag opens with a single headline level.

Universal labels, field glosses.
The same labels apply to every arXiv category. Fields do not add labels; they only state briefly what each label means in their context (Section 6).

Separate where from how much.
Saying in which section of a paper AI was used can mean very different things: "AI wrote the appendix" may describe a routine calculation, or a proof of a central theorem on which the main results rely. We argue that what matters is the *area* of the work AI contributed to (writing, proving, coding) and *how much* it contributed, recorded separately. Where classifications exist, they often combine the two; the STM classification [13], for example, lists language refinement and drafting as separate activities. We therefore use one label per area, wherever in the paper the contribution appears, each with its own depth: an AI-written appendix with an AI-generated proof becomes, for example, `Proof3 Text3`.

Short by default.
Labels at depth 0 are omitted, so a typical tag has two or three entries. A bare `AI-0` means no AI use at all, whereas `AI-0: Text1 Check2` records light supporting use that leaves the research itself untouched.

Distinguish only what matters.
The scheme should have as many labels and depths as necessary, but as few as possible, for two reasons. First, authors will use it only if it is quick and easy to apply, which in practice means a few minutes per paper rather than a long reporting form. Second, coarse categories are less ambiguous: the finer the distinctions, the more often two honest authors would classify the same work differently, and the more edge cases arise that require debate and a careful reading of the definitions. A distinction therefore earns its own label or depth only if it would change how a reader, referee, editor or moderator treats the paper. Whether AI checked a key computation or produced the proof clearly matters (`Check2` versus `Proof3`); whether it suggested a better word or a better paragraph structure does not, and both are recorded as `Text1`. Likewise, the scheme records that AI wrote analysis code (`Code2`), but not which programming tasks it helped with. Every label passes this test. The content labels each answer a different question about credit: who asked the question (`Idea`), who found the approach (`Method`), who formulated what is claimed (`Result`), and who established that it holds (`Proof`). The supporting labels each call for a different kind of scrutiny: AI-written prose for changes of meaning, AI-gathered references for invented citations, AI-written code and data for reproducibility, and AI-made figures for fabricated or misleading images. Labels that do not apply cost nothing, since they are omitted at depth 0.

Provenance, not purity.
Tags describe what happened. They do not rank papers, and higher depth is not a mark of lower quality. Light use in text and verification is good practice and should be reported as such (Section 3).

Machine-readable.
The tag should live in metadata, with a fixed grammar, so that it can be searched and aggregated.

Human responsibility is constant.
Whatever the tag says, the authors take full responsibility for the content. This is a rule of the scheme, not a part of the tag.

5 The scheme

5.1 Labels

Table 2 lists the ten labels. The first four carry the intellectual content of a paper and primarily determine its content level; the remaining six describe the work that supports it: its presentation (`Text`, `Figs`), its sources (`Lit`), its implementation and data (`Code`, `Data`), and its verification (`Check`).

Table 2: Labels, applicable to all fields. Field-specific readings are given in Table 4.

Label	Scope
Content labels	
`Idea`	Research questions, hypotheses, conjectures.
`Method`	Approach, design of the study, choice of model, proof strategy.
`Result`	Formulation of the main results: their statement, sharpening or generalization.
`Proof`	Justification of the results: proofs, derivations, analysis and interpretation of evidence.
Supporting labels	
`Text`	Prose, language, structure of the write-up, LaTeX markup and cross-references.
`Lit`	Literature search and summary, citations, bibliography.
`Code`	Software, simulations, symbolic or numerical computation.
`Data`	Collection, cleaning, labeling, synthetic data.
`Figs`	Plots, diagrams, tables.
`Check`	Verification of computations, proofs, code, references.

5.2 Depth scale

Each label used is given a depth from 0 to 4 (Table 3).

Table 3: Depth of AI usage within a label.

Depth	Name	Meaning
0	None	No AI use (the default; omitted from the tag).
1	Polishing	Minor assistance; the human-produced content is essentially unchanged.
2	Directed	AI drafted the material under close human direction, and every step was checked by the authors.
3	Substantive	AI made a substantive contribution that the authors guided and verified but did not fully determine.
4	Autonomous	AI produced the material essentially autonomously; human input was limited to the initial prompt and final review.

To reduce ambiguity, each level is defined by a test rather than a feeling: level 1 does not change the meaning of any statement; level 2 is material the authors could have produced in the same form without AI, given time; level 3 contains at least one essential idea the authors did not supply; level 4 involves no essential human idea beyond the task specification.

Each level thus adds one question that can be answered yes or no, and five levels are where such questions run out. A finer scale, say from 0 to 9, would invite disagreement about neighboring steps that no test could settle; conversely, a coarser one, whether with three steps (none, some, much) or just one level fewer, would merge cases that should be treated differently, such as polishing and drafting, or guided and autonomous contributions (Section 3).

5.3 Content level

Every tag opens with a headline, written `AI-0` to `AI-4`. It answers, in one symbol, the question most readers ask first: *how much of the research itself came from AI?* We define the headline as the maximum of

- the highest depth among the four content labels `Idea`, `Method`, `Result` and `Proof`, and
- the highest depth among the supporting labels `Text`, `Lit`, `Code`, `Data` and `Figs`, minus 2.

Heavy supporting use should be reflected in the headline, because substantial drafting is rarely neutral: when AI writes most of a paper, it typically adds explanations, fills gaps in arguments, and makes choices of emphasis that the authors may not have intended or worked out themselves. There is also a question of fair comparison: heavy AI use in supporting areas can save weeks of work, for example in writing up, coding or preparing data, and a researcher who saves that time produces more in the same period than a colleague who does not. This advantage should be visible. It need not be discouraged, however. Tasks that take much time but little expertise are exactly those where automation lets researchers spend more of their time, and of the public funding behind it, on the research itself; disclosure simply keeps comparisons fair. Concretely, depth 3 in any supporting label gives at least `AI-1`, and depth 4 at least `AI-2`. A paper written up entirely by AI, from human ideas and proofs, is thus `AI-2`, not `AI-0`. Supporting work alone never raises the headline above `AI-2`, so the line of Section 3, which lies between `AI-2` and `AI-3`, is crossed only by essential contributions to the research itself; a paper whose main theorem was proved by AI is `AI-3` or `AI-4`.

`Check` is deliberately excluded from this rule. Verification adds confidence rather than content, and Section 3 encourages it at every depth. Should checking lead to stronger or corrected results, for example because the AI notices that a theorem holds in a stronger form, this is recorded under the content labels (here `Result`), not `Check`.

5.4 Syntax

A tag consists of the content level, a colon, and the labels with their depths, listed in the order of Table 2 and omitting depth 0. Two optional annotations may follow: `+Lean` (or `+Rocq`, formerly `+Coq`, `+Isabelle`, and so on) if the main results have been formally verified, and the model and date in square brackets. Numerical or symbolic checks, for example in Python, are recorded under `Check` instead. For example:

- `AI-0` — no AI use at all;
- `AI-0: Text1 Check2` — proofreading and AI-checked computations, the encouraged case;
- `AI-1: Method1 Text1 Lit2 Code3 Check2` — an empirical paper with human design and substantial AI-written analysis code;
- `AI-3: Proof3 Text1 Check2 +Lean [model, 2026-09]` — a mathematics paper with a human conjecture, a proof produced jointly with AI, and a formal certificate.

The model–date annotation is optional but recommended. AI capabilities change quickly, and `Code3` in 2026 will not mean the same as `Code3` some years later.

5.5 Assisted classification

Authors need not classify their papers unaided. We recommend that, at the end of the work, the author gives the AI system used a short excerpt defining the scheme (Appendix A), together with the paper or a description of the work, and asks it to propose a tag. The procedure is:

1. the AI proposes a depth for each label, with a one-line justification naming the specific contributions it made;
2. the author reviews the proposal and accepts or changes it;
3. if the author lowers a depth, noting the reason briefly in the paper's AI usage statement is recommended: lowering is often legitimate, for instance when an idea came from an earlier discussion the AI could not see, and a stated reason shows that the tag was considered rather than adjusted.

The value of this step is that the AI acts as a *witness*. Within a working session it has a record of which ideas, arguments and passages it supplied, whereas authors naturally come to regard the final version as their own. Its proposal is anchored in the written depth tests rather than in memory, and it turns disagreements into concrete questions, such as whether a particular idea was essential. It also lowers the barrier to adoption: classifying a paper this way should only take a few minutes.

The AI is not, however, a neutral judge. It can overstate its contribution, since it does not know what the authors had already worked out alone or with colleagues, and it can understate it, since language models tend to defer to their users and may lower a depth under pressure. It usually sees only the sessions in which it was used, not work done with other tools, with human colleagues, or by the author alone. For papers with several authors, each of whom may have used their own AI, a classification from only one author's sessions under-reports by design; each author can then classify their own share, and the tag takes the highest depth per label. Assisted classification therefore remains a form of self-reporting, and the author remains responsible for the final tag. It works best when carried out at the end of the session in which the work was done, or from saved logs, and when the AI is first asked for its own proposal before the author comments on it. Authors should also check that uploading unpublished work is consistent with the rules of their institution and of the intended journal, and agreed with their co-authors; the procedure is recommended, not required.

6 Field glosses

The labels are the same in every field; what changes is what a "result," a "proof," or a "method" typically is. Table 4 gives short glosses for the main arXiv groups. They are intended as reading aids, not as additional categories. `Idea` needs no gloss: its scope (research questions, hypotheses, conjectures) already covers every field.

Table 4: Field glosses for the content labels.

Field	`Result`	`Proof`	`Method`
Mathematics, theoretical CS	theorems, lemmas, conjectures settled	proofs	proof strategy, choice of approach
Physics	predictions, derived laws, computed quantities	derivations, calculations	approximations, parameter regimes, boundary conditions
CS and machine learning	algorithms, guarantees, empirical claims	proofs of guarantees, experimental evidence	architectures, experimental design
Statistics, q-bio, q-fin, economics	findings, estimates	analysis and interpretation of evidence	study design, model specification
Electrical engineering, systems	designs and their performance	analysis, simulation evidence	design choices for circuits, controllers, layouts

A few points need more than a gloss.

Mathematics. The distinction between `Result` and `Proof` is the heart of the scheme for mathematics: it separates an AI proof of a human conjecture (`Proof3`) from a human proof of an AI conjecture (`Idea3`), and both from AI sharpening of a human theorem (`Result3`). Formal verification is recorded with `+Lean` or similar, and AI assistance in producing the formalization under `Check`. An AI proof of a routine lemma, one the authors could have supplied and would otherwise have omitted as easy, is `Proof2`: it adds checked detail, not an essential idea.

Physics. Approximations, parameter regimes, and boundary conditions can determine results while being invisible in the final text. If AI chose them, this belongs under `Method`.

Computational work. Where the code *is* the method, as in a simulation study whose central contribution is the numerical scheme, AI contributions to its design are recorded under `Method`, and only the implementation under `Code`.

Computer science and machine learning. Many papers in these fields study AI systems. AI as the *object* of research must be distinguished from AI as a *tool*; otherwise every machine-learning paper would appear to involve heavy AI use. The tag records only tool use. A paper training a language model and evaluating it scores nothing for that; a paper whose evaluation code was written by an AI assistant records it under `Code`.

Empirical fields. Reproducibility risk concentrates in `Data` and in the choice and interpretation of analyses. We recommend that `Data2` or higher, or `Method3` or higher, be accompanied by what is needed to reproduce the AI's part: the scripts, or the prompts where data were generated by prompting. This serves reproducibility rather than attribution, and differs from publishing full chat logs (Section 7).

7 Issues and limitations

Self-reporting. The scheme relies on authors' honesty and judgment. Tests attached to each depth level (Section 5) reduce, but do not remove, variation between authors. Worked examples and community guidance will be needed, as they are for subject classifications, and agreement should improve as authors gain experience with the scheme and its conventions settle. Assisted classification (Section 5.5) adds a contemporaneous record of what the AI contributed, but does not remove the author's judgment or responsibility.

Detection. Under Article 50(2) of the EU AI Act, providers of generative AI systems must, from August 2026, mark their outputs, including text, so that they are detectable as AI-generated; a Code of Practice sets out how [14]. Signing the Code is voluntary; as of September 2026, its signatories include Anthropic, Google, Meta, Microsoft, Mistral and OpenAI, but not xAI, the provider of Grok [15]. For text, this is done by statistical watermarking: a secret key steers the model's choice among plausible words, invisibly to readers but detectably for the key holder [16, 17]. Such marks could complement self-reporting, but they measure only who chose the words, that is, `Text`, a supporting label. An AI-generated proof written up by the authors carries no mark, whereas an AI translation of an entirely human paper is marked throughout. A detected watermark is therefore a statistical signal about the write-up, not about the research itself, and a missing one proves nothing.

An idealized distinction. The scheme separates the authors' own contribution from that of AI more sharply than research practice does. Ideas arise from interaction with colleagues, the literature and, increasingly, AI, and the boundary between them is rarely clear-cut. The labels and depths are therefore a convention for reporting, comparable to the norms for authorship and citation, not an empirical claim that contributions can be separated exactly.

Prompts and logs. Some proposals ask authors to publish their prompts and the AI's replies. We do not require this. Ideas develop across many sessions, tools and conversations, just as they once grew from a book, an article and a discussion with a colleague, so a selection of chat logs would be incomplete as well as laborious to compile. A requirement to publish them would also reward reporting lower depths to avoid the effort, and logs may contain personal information or unpublished ideas that authors are entitled to withhold. Logs are most informative where AI did most of the work: for contributions at depth 4, the prompts show what direction the authors gave, and publishing them is encouraged, as are the prompts or scripts behind AI-generated data and analyses in empirical fields (Section 6).

Thresholds for depth 0. Spell-checkers, autocomplete, and AI-assisted search engines are now ubiquitous. We propose that conventional tools, such as a dictionary-based spell-checker or a keyword search engine, do not count. Any tool that rewrites, generates, summarizes, or checks content counts, and is reported at the appropriate depth; for proofreading and checking this will typically be `Text1` or `Check1`–`Check2`, which Section 3 treats as encouraged.

Incentives. If any tag carries stigma, authors will under-report. The scheme must therefore be presented, and used by readers, referees, editors and moderators, as a statement of provenance and not a quality score, with light use explicitly treated as good practice (Section 3). Conversely, where AI contributions attract attention, for example in showcases of AI-generated results, over-reporting may appear; the formal-verification annotation and the availability of prompts mitigate this by tying claims to checkable facts.

Gaming the headline. An author might be tempted to report an essential AI contribution under `Text` or `Code` to keep the headline low. Because heavy supporting use raises the headline (Section 5), this gains at most one or two levels and never hides an essential contribution entirely. The depth tests and field glosses are meant to prevent such misreporting, and referees can query a tag that does not match the paper. Assisted classification also helps: where the AI's proposal is kept on record, a depth lowered without stated reason is visible as such. The risk nonetheless remains.

Granularity. A single tag describes a whole paper. Papers with several main results of different provenance may give a tag per result in the body, with the metadata tag giving the maximum depth per label.

Non-standard papers. Not every paper has theorems or experiments. For a survey, position paper or proposal such as this one, the `Result` is the proposal itself and the `Proof` is the argument made for it. Some borderline cases also need a convention: we record LaTeX source under `Text`, not `Code`, since it is part of the write-up. Similarly, a TikZ diagram drawn from data with labeled axes is coded, represents data and contains words, but it is recorded once, under `Figs`: a reader wants to know whether the figures were made by AI, and a single label answers exactly that. Likewise, a reference found by AI that supplies a key step, such as a lemma whose applicability the AI asserted, counts as a contribution to `Method` or `Proof`, not only `Lit`: checking that such a result really applies is among the most error-prone parts of AI-assisted work.

Versioning. Tags should be updated with each version of a preprint. A second version that incorporates AI-found corrections, for example, would add or raise `Check`.

Scope. The scheme records AI use; it does not judge whether to use AI at all. Some researchers decline AI for ethical reasons, either entirely or from particular companies. Access to the strongest models and to compute is also unequal, for financial and institutional reasons and between countries. Disclosure makes such differences visible but does not resolve them.

8 Comparison with existing proposals

Several frameworks for describing contributions to research, or AI use in particular, already exist. Table 5 compares them with the present proposal on five features.

Contributor roles. The Contributor Roles Taxonomy (CRediT) [11, 18], standardized as ANSI/NISO Z39.104-2022, describes *human* contributions through roles such as conceptualization, formal analysis, software and writing, each with a degree of lead, equal or supporting. Our labels deliberately align with CRediT roles where possible, and one route to adoption is to extend CRediT to non-human contributors.

Publishing taxonomies. The STM Association's classification of AI use in manuscript preparation [13] distinguishes nine activities, such as language refinement, drafting, translation, data visualization and reference gathering, and states for each whether disclosure is expected. It explicitly excludes the use of AI in the research process itself, which is where disclosure matters most. The Generative AI Delegation Taxonomy (GAIDeT) [19] covers the whole research cycle by combining research stages with the roles AI played, and generates a standard declaration text. AI Usage Cards [20] document AI use per stage (ideation, methods, writing, code and data, ethics) in a structured card with machine-readable export, marking content as new, revised or compared. These frameworks are broad, but none grades the *depth* of AI involvement, and their output is a statement or card rather than a tag that can be read at a glance.

Compact statements and levels. IBM's AI Attribution Toolkit [21] produces short coded statements, modeled on Creative Commons licenses, describing the balance of human and AI effort, what AI contributed, and who reviewed the work; it is aimed at creative work in general rather than research. In education, the AI Assessment Scale [22] uses five levels of permitted AI use. For mathematics, Feng et al. [23] classify AI-assisted results by autonomy, as primarily human (H), human–AI collaboration (C) or essentially autonomous (A), alongside a separate scale of mathematical significance, and propose human–AI interaction cards documenting the exchanges behind a result. Also for mathematics, Hirschi [24] has proposed a complementary classification of papers by the degree of human understanding they retain, building on an earlier version of the present proposal. The Hexagon repository [8], created in response to the changes AI has brought to mathematical research, also accepts works that authors may not wish, or be able, to submit to arXiv. It labels each deposit by who wrote the text (primarily AI-generated, mixed, or human-written) and by how much of the work humans understand (all, some or no parts). Like Hirschi's classification, these labels record the human side of a result and complement the present scheme: the first corresponds to our `Text`, while human understanding has no counterpart in AIL2026.

Principles. The Leiden Declaration on Artificial Intelligence and Mathematics [25], endorsed by the International Mathematical Union [26], calls on mathematicians to disclose their use of AI carefully and keeps AI out of authorship. It recommends a dedicated disclosure section in papers, but prescribes no taxonomy or format for it. A global reporting standard for AI

disclosure is being developed jointly by STM, COPE, the International Science Council and the Global Young Academy [27]; its consultations ask precisely which uses should be disclosed, where, and with what taxonomy.

Table 5: Comparison of frameworks. Columns: records use *by area*; *graded* depth of involvement; *compact* tag; single *headline* level; *separates* ideas, results and proofs. ✓: yes; ○: partly; –: no.

Framework	Scope	By area	Graded	Compact	Headline	Separates
CRediT [11]	human contributor roles	✓	○	–	–	–
STM [13]	manuscript preparation only	✓	–	–	–	–
GAIDeT [19]	research and publication	✓	–	–	–	○
AI Usage Cards [20]	research and writing	✓	○	–	–	○
AI Attribution [21]	creative work in general	○	○	✓	–	–
Feng et al. [23]	AI-generated mathematics	–	✓	✓	✓	–
Hexagon [8]	mathematics repository	○	✓	✓	–	–
Leiden Declaration [25]	mathematics, principles	–	–	–	–	–
This proposal	all arXiv fields	✓	✓	✓	✓	✓

Advantages of the present proposal. The scheme combines features that are currently spread across separate frameworks:

- *Readable at a glance.* The headline content level answers the first question most readers ask, and the labels are plain words. No other framework offers both a single headline and a breakdown by area.
- *Breadth and depth together.* It records *where* AI was used, as GAIDeT and AI Usage Cards do, and *how deeply*, as the levels of Feng et al. and the AI Assessment Scale do, in a single short string.
- *The research itself, not only the manuscript.* Unlike the STM classification, it covers ideas, methods, results and proofs.
- *The distinctions mathematicians care about.* Separating `Idea`, `Result` and `Proof` distinguishes an AI proof of a human conjecture from a human proof of an AI conjecture, which a single autonomy level cannot.
- *Encouragement, not only disclosure.* It treats light use, and AI checking in particular, as good practice, and draws its line on the explicit basis of fairness (Section 3).
- *Metadata, not prose.* Like MSC2020, it is designed to sit in a paper's metadata, where it can be searched and aggregated.

The proposal is complementary rather than competing. It offers a concrete format for the disclosure the Leiden Declaration calls for, gives one answer to each of the questions posed by the global reporting-standard consultation, and can serve as a compact summary of a fuller statement such as an AI Usage Card or a set of human–AI interaction cards.

Trade-offs. The compactness has costs. A tag conveys less than a card or an interaction log: it says how much AI contributed, not what was asked or answered, so fuller documentation remains valuable for papers with high content levels. Graded depths require more judgment than the yes/no categories of STM or GAIDeT, and agreement between authors has yet to be measured. Finally, any new scheme adds to an already crowded field; it will succeed only if it aligns with whatever standard publishers and preprint servers adopt.

9 Path to adoption

1. **Pilot in mathematics.** The community is relatively small, has experience with a shared classification (MSC), and is directly affected by recent advances in AI-generated

proofs [4, 5, 6]. A pilot could begin with volunteer authors adding the tag, with a reference to its definitions, to their arXiv comments field.

2. **Publish guidance and examples.** A short guide with worked examples of each depth level for each label, together with the classification excerpt and prompt of Appendix A for assisted classification.
3. **Gather feedback and revise.** Measure agreement between authors and between authors and referees on a sample of papers, and adjust definitions where agreement is low.
4. **Contribute to emerging standards.** Submit the scheme to the consultation on a global AI-disclosure reporting standard [27], as a compact encoding compatible with it.
5. **Propose a metadata field.** Once the scheme is stable, propose an optional arXiv metadata field alongside the MSC and ACM classes, and invite journals to adopt it.
6. **Offer a tag assistant.** A short questionnaire, or an AI that asks a few questions about the work, could propose a tag during submission. It should help, not monitor: its suggestion should not be stored or compared with the final tag, so that authors stay in control and have no reason to answer strategically.

10 Conclusion

Free-text disclosure of AI use is better than none, but it is not enough. AIL2026, a short tag of plain-word labels and depths, headed by a single content level, would make AI use visible, searchable, and comparable, and would make human contributions clearer at the same time. It should encourage the light uses that improve papers, such as proofreading and checking, while making sure that essential AI contributions are never presented as unassisted. We invite comment from the mathematical community and beyond, and propose a pilot as the next step.

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AI usage of this paper

As a worked example, we classify this paper itself. Following Section 5.5, the tag was proposed by the AI system used, on the basis of the working session, and reviewed and accepted by the author. It is

```
AI-3: Idea3 Result3 Proof3 Text3 Lit2 Figs3 Check2 [Claude Opus 5.5, 2026-10]
```

Following Section 7, the `Result` of a proposal such as this is the proposal itself and the `Proof` is the argument made for it.

`Idea3` The core idea is the author's: a label-based classification of AI use in scientific articles, allowing readers to grasp at a glance in which ways and to what depth AI has contributed to the findings and their presentation. It further includes the encouragement of light use, such as text polishing and result checking, which benefits the whole community, and the fairness principle, which entails disclosing how AI has accelerated the authors' workflow. The author also suggested providing a prompt that lets the AI used propose an initial tag itself. AI contributed further essential ideas, such as the treatment of verification, the

distinction between AI as object and as tool, the model–date annotation, and the relation and comparison to existing proposals.

`Result3` The final scheme was designed jointly. The author set its requirements (universal labels, readable at a glance); AI proposed the specific labels, the depth tests, and the headline content level.

`Proof3` The central arguments of Section 3 are the author's; AI drafted them and supplied further arguments, notably in the motivation, the design principles and the discussion of limitations.

`Text3` AI drafted the prose and LaTeX source under the author's direction, and the author reviewed and edited it.

`Lit2` AI suggested the initial reference points and, at the author's request, searched for and summarized the related proposals of Section 8 and located and checked sources for the recent developments cited in the introduction.

`Figs3` AI designed all tables.

`Check2` AI compiled the source, checked that all cross-references resolve and the tables render correctly, and, at the author's request, checked all references against their original sources and corrected errors.

The author has reviewed all AI-generated content, including the corrected references, and takes full responsibility for the content. This tag describes the current version and will be updated with future versions.

A Classification excerpt and prompt

The following excerpt of AIL2026 and the prompt are intended to be copied, together with the paper or a description of the work, into the AI system that was used.

A.1 Excerpt

Labels. *Content labels:* `Idea` (research questions, hypotheses, conjectures); `Method` (approach, study design, choice of model, proof strategy); `Result` (formulation of the main results); `Proof` (proofs, derivations, analysis and interpretation of evidence). *Supporting labels:* `Text` (prose, language, LaTeX markup and cross-references); `Lit` (literature search, citations, bibliography); `Code` (software, computation); `Data` (collection, cleaning, synthetic data); `Figs` (plots, diagrams, tables); `Check` (verification of computations, proofs, code, references).

Depths.

0 no AI use (omitted from the tag);
1 polishing: does not change the meaning of any statement;
2 directed: AI drafted under close human direction, and the authors could have produced the same material without AI, given time;
3 substantive: contains at least one essential idea the authors did not supply;
4 autonomous: no essential human idea beyond the task specification.

Content level. `AI-n`, where n is the maximum of the highest depth among `Idea`, `Method`, `Result` and `Proof`, and the highest depth among `Text`, `Lit`, `Code`, `Data` and `Figs` minus 2. `Check` does not count: AI verification is encouraged at every depth, and stronger or corrected results found by checking are recorded under the content labels. Levels `AI-3` and `AI-4` mean that AI contributed essential parts of the research.

Syntax. `AI-n: Label<d> Label<d> ... [+Lean] [model, YYYY-MM]`, with labels in the order above and depth 0 omitted. For proposals, position papers and surveys, `Result` is the proposal and `Proof` the argument for it. LaTeX source counts as `Text`.

A.2 Prompt

```
You have assisted with the work described in the attached paper.
Using the classification excerpt above, propose an AI usage tag.

1. For each label, assign a depth from 0 to 4 by applying the
   depth tests as written. Base each depth only on contributions
   you can identify in this session or in the material provided,
   and name them in one line.
2. Give the content level: the maximum of the highest depth among
   Idea, Method, Result and Proof, and the highest depth among
   Text, Lit, Code, Data and Figs minus 2. Check does not count.
3. State which parts of the work you cannot see (other sessions,
   other tools, human collaborators) and where you are uncertain.
4. Do not raise or lower a depth to please the author. If the
   author disagrees, explain how the definition applies; the
   author makes the final decision.

Output the tag in the form
   AI-n: Label<d> Label<d> ... [model, YYYY-MM]
followed by a one-line justification for each label.
```

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